

Supplementary File S4

RQ4: Occupant Framing, Human-Centric Metrics, and Metric Imbalances

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ4. It reports occupant-framing scores, human-centric metric visibility, metric imbalances, gap signals, and concise survey-level answers.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

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S4.1 The reinforcement learning method for occupant behavior in building control: A review [1]

Table S4.1: RQ4 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Occupant-centered scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Han et al. (2021)	3	3	3	2	3	The paper is a specialized review of reinforcement learning for occupant behavior in building control. Occupant-centered scope receives score 3 because occupant behavior is the declared and structuring focus of the review. Occupant-framing visibility receives score 3 because occupants are represented not only as sources of uncertainty, but also through presence, activity, comfort feedback, state information, state-transition influence, reward-related disturbance, and feedback input. Metric visibility receives score 3 because comfort level and energy saving are explicitly identified as the two main reward types in the reviewed studies. Metric-imbalance visibility receives score 2 because the energy–comfort structure is clear, but the paper is not mainly organized as a metric-imbalance review. Gap-signal visibility receives score 3 because the paper explicitly identifies limitations around using the same comfort criterion for different individuals, assuming constant occupant behavior, and the relative scarcity of lighting, window, and vessel controls.

Table S4.2: RQ4 occupant-framing scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Han et al. (2021)	2	3	2	3	2	1

Table S4.3: RQ4 metric-visibility scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Han et al. (2021)	3	1	0	3	1	2	3	2

Table S4.4: RQ4 gap-signal scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Han et al. (2021)	3	2	1	2	1

Table S4.5: RQ4 summary table for Han et al. (2021), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Han et al. (2021)	Mixed, but energy–comfort dominant	The review conceptualizes occupants as stochastic and behaviorally important actors who enter RL control as uncertainty sources, comfort-constrained users, feedback providers, and explicit behavior-related elements within the MDP. However, the strongest recurring empirical pattern remains the use of occupant-related information to support HVAC-oriented control under an energy–comfort trade-off.	For RQ4, the paper is best interpreted as evidence that occupant framing in RL-for-buildings can be richer than simple comfort-constrained optimization, because occupants appear as states, state-transition influences, reward modifiers, and feedback sources. Even so, the metric language remains centered on energy saving and thermal comfort, while personalization, richer human feedback, and broader non-HVAC human-centric control remain less developed.

Table S4.6: One-line RQ4 answer for Han et al. (2021), after recoding.

Survey	RQ4 answer
Han et al. (2021)	Han et al. (2021) portrays occupants as stochastic but behaviorally relevant actors who can enter RL-based building control as uncertainty sources, comfort-constrained users, feedback providers, and explicit behavioral elements, yet the reviewed literature still remains largely structured around an energy–comfort trade-off, with richer personalization and broader human-centric control applications less developed.

S4.2 Systematic review on deep reinforcement learning-based energy management for different building types [2]

Table S4.7: RQ4 recoded audit table for Shaqour & Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad BEMS scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Shaqour and Hagishima (2022)	3	2	3	3	2	The paper is a broad, building-type-centric review of DRL-based BEMS. Broad BEMS scope receives score 3 because the review is structured around BEMS applications across different building types. Occupant-framing visibility receives score 2 because occupants appear through comfort requirements, health-related indoor-environment constraints, human presence, and behavior as stochastic factors, but occupants do not organize the whole review. Metric visibility receives score 3 because the paper explicitly covers energy consumption, energy cost, demand response, human comfort, health constraints, discomfort, productivity, PM2.5, and indoor CO ₂ . Metric-imbalance visibility receives score 3 because the paper makes the dominance of energy, cost, demand-response, renewable-integration, and efficient-control objectives visible while human-centric concerns appear mainly as secondary comfort and health layers. Gap-signal visibility receives score 2 because real-world validation and human-centric validation limitations are discussed, but occupant-specific gaps do not structure the entire paper.

Table S4.8: RQ4 occupant-framing scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Shaqour & Hagishima (2022)	2	3	0	1	0	2

Table S4.9: RQ4 metric-visibility scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Shaqour & Hagishima (2022)	3	3	3	2	2	0	1	0

Table S4.10: RQ4 gap-signal scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Shaqour & Hagishima (2022)	0	0	1	1	2

Table S4.11: RQ4 summary table for Shaqour & Hagishima (2022), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Shaqour & Hagishima (2022)	Mostly energy-centric, but clearly multi-objective	The review mainly frames occupants through comfort requirements, health-related indoor-environment constraints, and human presence or behavior as stochastic factors that shape BEMS design and operation. It gives little direct attention to occupants as explicit behavioral models, feedback-giving agents, or personalization targets.	For RQ4, the paper is best interpreted as showing that DRL-based BEMS research is organized primarily around energy, cost, and demand-response objectives, while human-centric concerns enter mainly as comfort, discomfort, productivity, and health constraints. Thus, the review supports a clear metric imbalance in which occupant-centered outcomes are present and meaningful, but remain secondary to broader energy-management objectives.

Table S4.12: One-line RQ4 answer for Shaqour & Hagishima (2022), after recoding.

Survey	RQ4 answer
Shaqour and Hagishima (2022)	Shaqour & Hagishima (2022) portray occupants mainly as comfort- and health-constrained users whose presence and behavior add stochasticity to BEMS control, but the surveyed literature remains mostly organized around energy, cost, and demand-response objectives, with human-centric metrics such as discomfort, productivity, PM2.5, and indoor CO ₂ appearing as important but secondary objective layers.

S4.3 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [3]

Table S4.13: RQ4 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	RES/BEMS scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Michailidis et al. (2025)	3	2	2	3	1	The paper is a specialized review of reinforcement learning for optimizing renewable energy utilization in buildings. RES/BEMS scope receives score 3 because renewable-energy utilization in BEMS is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because the paper includes occupant presence and occupant behavior as part of the dynamic BEMS environment and repeatedly treats occupant comfort as a co-objective, but occupants do not structure the review. Metric visibility receives score 2 because occupant comfort appears in multi-objective reward functions and target-output discussions, but energy efficiency, cost reduction, renewable utilization, peak demand, self-consumption, storage management, and grid interaction remain more visible. Metric-imbalance visibility receives score 3 because the paper clearly shows comfort as secondary to energy-management and RES-integration objectives. Gap-signal visibility receives score 1 because behavioral realism and real-life validation issues appear weakly, but occupant-specific gaps are not strongly developed.

Table S4.14: RQ4 occupant-framing scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Michailidis et al. (2025)	2	2	0	1	0	0

Table S4.15: RQ4 metric-visibility scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Michailidis et al. (2025)	3	3	2	2	0	0	1	0

Table S4.16: RQ4 gap-signal scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Michailidis et al. (2025)	0	0	0	1	1

Table S4.17: RQ4 summary table for Michailidis et al. (2025), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Michailidis et al. (2025)	Mostly energy-centric, but multi-objective	The review mainly frames occupants as comfort-constrained users and as part of the dynamic environment through occupant presence and occupant behavior. It gives little direct attention to occupants as explicit behavioral models, preference-feedback sources, or personalization targets.	For RQ4, the paper is best interpreted as showing that RL research on RES-integrated BEMS is organized primarily around energy efficiency, cost reduction, renewable utilization, storage management, and grid-related performance, while occupant comfort appears as a recurring co-objective inside broader multi-objective reward structures. Thus, the review supports a clear imbalance in which human-centric metrics are present but secondary to energy-management objectives.

Table S4.18: One-line RQ4 answer for Michailidis et al. (2025), after recoding.

Survey	RQ4 answer
Michailidis et al. (2025)	Michailidis et al. (2025) portray occupants mainly as comfort-constrained users embedded in a dynamic building environment, while the reviewed RL literature for RES-integrated BEMS remains mostly organized around energy efficiency, cost reduction, renewable self-consumption, storage management, and grid interaction, with occupant comfort treated as an important but secondary co-objective rather than as the dominant survey focus.

S4.4 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [4]

Table S4.19: RQ4 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	HVAC scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Al Sayed et al. (2024)	3	2	3	2	2	The paper is a specialized review of reinforcement learning for HVAC control in intelligent buildings. HVAC scope receives score 3 because HVAC control is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because occupants appear mainly through comfort requirements, indoor air quality and CO ₂ -related outcomes, and occupancy schedules as exogenous state components, but occupants do not structure the paper as behavioral agents. Metric visibility receives score 3 because energy, comfort, cost, and air-quality-related objectives are repeatedly visible in the HVAC formulation and reward discussion. Metric-imbalance visibility receives score 2 because the energy–comfort framing is clear, but metric imbalance is not itself the main organizing analysis of the paper. Gap-signal visibility receives score 2 because the paper discusses poor generalization to real-world variation in exogenous state components, including occupancy schedules, electricity prices, and weather, together with limited real-building deployment; however, these are broader deployment and generalization gaps rather than specifically occupant-centered gaps.

Table S4.20: RQ4 occupant-framing scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Al Sayed et al. (2024)	2	3	0	0	0	2

Table S4.21: RQ4 metric-visibility scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Al Sayed et al. (2024)	3	2	1	3	2	0	1	0

Table S4.22: RQ4 gap-signal scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Al Sayed et al. (2024)	0	0	1	2	2

Table S4.23: RQ4 summary table for Al Sayed et al. (2024), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Al Sayed et al. (2024)	Mixed, but energy–comfort dominant	The review mainly frames occupants as comfort-constrained users whose presence is represented indirectly through exogenous state components such as occupancy schedules, while air quality and CO ₂ appear as additional indoor-environment concerns. It gives little direct attention to explicit behavioral modeling, preference feedback, or personalization.	For RQ4, the paper is best interpreted as showing that RL-for-HVAC research is organized primarily around the energy–comfort trade-off, with cost minimization and some air-quality-related objectives added on top. Thus, the review supports a clear imbalance in which human-centric outcomes are present and important, but remain embedded within a predominantly HVAC-optimization and energy-management framing rather than forming a richly occupant-centered survey logic.

Table S4.24: One-line RQ4 answer for Al Sayed et al. (2024), after recoding.

Survey	RQ4 answer
Al Sayed et al. (2024)	Al Sayed et al. (2024) portray occupants mainly as comfort-constrained users represented through indoor-environment objectives and exogenous occupancy-related conditions, while the reviewed RL-for-HVAC literature remains largely structured around an energy–comfort trade-off with additional cost and air-quality terms rather than around explicit behavioral modeling, feedback-rich interaction, or personalization.

S4.5 Reinforcement learning for building controls: The opportunities and challenges [5]

Table S4.25: RQ4 recoded audit table for Wang & Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad building-control scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Wang and Hong (2020)	3	2	3	3	3	The paper is a broad review of reinforcement learning for building controls. Broad building-control scope receives score 3 because the paper is framed as a comprehensive review of RL for building-control applications. Occupant-framing visibility receives score 2 because occupants are discussed through comfort, environmental adaptation, and feedback, but occupant behavior or personalization does not structure the whole review. Metric visibility receives score 3 because the paper structurally discusses multiple control objectives, including occupant comfort, energy conservation, grid flexibility, carbon reduction, and reward formulation. Metric-imbalance visibility receives score 3 because the review makes visible that the literature is much more developed around thermal comfort, energy conservation, and load flexibility than around broader human-centric metrics. Gap-signal visibility receives score 3 because limited occupant feedback, rare indoor air quality treatment, and rare visual-comfort treatment are strong human-centric gap signals in the review synthesis.

Table S4.26: RQ4 occupant-framing scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Wang & Hong (2020)	2	3	2	1	0	2

Table S4.27: RQ4 metric-visibility scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Wang & Hong (2020)	3	2	3	3	2	1	1	0

Table S4.28: RQ4 gap-signal scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Wang & Hong (2020)	0	3	3	1	1

Table S4.29: RQ4 summary table for Wang & Hong (2020), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Wang & Hong (2020)	Mixed, but energy–flexibility–comfort dominant	The review mainly frames occupants as comfort-constrained building users whose needs must be balanced against energy conservation and grid flexibility, while also recognizing occupants as part of the adaptive environment and, in principle, as sources of feedback for human-in-the-loop control. However, explicit behavioral or personalized occupant modeling remains limited.	For RQ4, the paper is best interpreted as showing that the RL-for-building-controls literature acknowledges occupant comfort and occupant feedback conceptually, but in practice remains much more developed around thermal comfort, energy conservation, and load flexibility than around richer human-centric dimensions such as indoor air quality, visual comfort, and systematic human-in-the-loop interaction.

Table S4.30: One-line RQ4 answer for Wang & Hong (2020), after recoding.

Survey	RQ4 answer
Wang and Hong (2020)	Wang & Hong (2020) portray occupants mainly as comfort-constrained users embedded in an adaptive control setting, with occupant feedback acknowledged as important but still rarely integrated in practice, while the surveyed RL literature remains largely organized around thermal comfort, energy conservation, and grid flexibility rather than richer human-centric metrics such as IAQ, visual comfort, and feedback-rich occupant interaction.

S4.6

Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [6]

Table S4.31: RQ4 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	MORL/smart-building scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Huynh et al. (2025)	3	2	3	3	3	The paper is a broad systematic review of multi-objective reinforcement learning in smart buildings. MORL/smart-building scope receives score 3 because MORL in smart buildings is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because occupants are discussed through user comfort, changing occupant needs, occupancy patterns, user feedback, complaints, and preference-aware tuning, but occupants do not structure the whole survey. Metric visibility receives score 3 because the paper structurally analyzes objectives and conflicts in MORL-based smart-building control, including energy consumption, energy cost, thermal comfort, user comfort, indoor environmental quality, and other building-management objectives. Metric-imbalance visibility receives score 3 because the objective-frequency and conflict analysis make visible that energy, cost, and thermal comfort dominate, while broader IEQ, visual comfort, acoustic comfort, and personalization appear less frequently. Gap-signal visibility receives score 3 because occupant behavior, feedback inconsistency, personalized comfort, and broader IEQ dimensions are discussed as challenge or future-direction issues.

Table S4.32: RQ4 occupant-framing scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Huynh et al. (2025)	2	3	2	1	1	2

Table S4.33: RQ4 metric-visibility scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Huynh et al. (2025)	3	3	2	3	2	2	2	1

Table S4.34: RQ4 gap-signal scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Huynh et al. (2025)	2	2	3	2	1

Table S4.35: RQ4 summary table for Huynh et al. (2025), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Huynh et al. (2025)	Mixed, but energy–cost–comfort dominant	The review mainly frames occupants as comfort-bearing users embedded in a dynamic smart-building environment shaped by occupancy patterns, changing needs, and, in some cases, explicit feedback or preference signals. However, occupants are not primarily treated as richly modeled behavioral agents, and personalization remains secondary.	For RQ4, the paper is best interpreted as showing that MORL research in smart buildings is organized most strongly around the joint optimization of energy consumption, energy cost, and thermal comfort, while user comfort and indoor environmental quality concerns are present but less dominant, and richer human-centered dimensions such as visual comfort, acoustic comfort, health-oriented air-quality objectives, and feedback-rich personalization remain comparatively thinner in the survey literature.

Table S4.36: One-line RQ4 answer for Huynh et al. (2025), after recoding.

Survey	RQ4 answer
Huynh et al. (2025)	Huynh et al. (2025) portray occupants mainly as comfort-bearing users whose needs, schedules, and occasional feedback shape MORL control, but the surveyed literature remains dominated by energy consumption, energy cost, and thermal comfort objectives, with broader human-centric outcomes such as IAQ, visual comfort, acoustic comfort, and personalization receiving much lighter attention.

S4.7 A review of reinforcement learning for autonomous building energy management [7]

Table S4.37: RQ4 recoded audit table for Mason & Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad BEM scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Mason and Grijalva (2019)	3	2	3	2	1	The paper is a broad review of reinforcement learning for autonomous building energy management. Broad BEM scope receives score 3 because the paper reviews RL applications across building energy-management domains, including HVAC, water heating, lighting, smart appliances, PV, batteries, EVs, and smart-grid interaction. Occupant-framing visibility receives score 2 because occupants appear mainly through thermal comfort and consumer behavior, especially in HVAC and appliance-related control, but occupants do not structure the review as behavioral agents. Metric visibility receives score 3 because energy use, energy cost, demand/grid interaction, scheduling, storage, and thermal comfort are visible across the reviewed application areas. Metric-imbalance visibility receives score 2 because the energy/cost orientation is clear while comfort appears mainly as a constraint or co-objective, but the paper is not organized as a metric-imbalance review. Gap-signal visibility receives score 1 because the multi-objective energy-cost/thermal-comfort trade-off is recognized as an important issue, but occupant-centered gaps such as personalization, feedback-rich interaction, and broader IEQ are not strongly developed as future-work themes.

Table S4.38: RQ4 occupant-framing scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Mason & Grijalva (2019)	1	2	0	0	0	1

Table S4.39: RQ4 metric-visibility scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Mason & Grijalva (2019)	3	3	2	2	1	0	1	0

Table S4.40: RQ4 gap-signal scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Mason & Grijalva (2019)	0	0	1	1	0

Table S4.41: RQ4 summary table for Mason & Grijalva (2019), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Mason & Grijalva (2019)	Mixed, but energy–cost dominant	The review mainly frames occupants as users whose thermal comfort must be maintained while reducing energy use and energy cost, especially in HVAC control. It gives only minor attention to broader occupant interaction, explicit behavioral representation, or personalization.	For RQ4, the paper is best interpreted as showing that early RL-for-building-energy-management surveys were already aware of the energy–comfort trade-off, but still treated human-centered concerns mainly through thermal comfort rather than through richer occupant modeling, feedback, or broader indoor-environment quality. Thus, the survey supports a clear energy-centric tendency with comfort acting mainly as a constraint or secondary co-objective.

Table S4.42: One-line RQ4 answer for Mason & Grijalva (2019), after recoding.

Survey	RQ4 answer
Mason and Grijalva (2019)	Mason & Grijalva (2019) portray occupants mainly as thermal-comfort-constrained users within an energy-management problem, while the reviewed RL literature remains organized much more strongly around energy savings, energy cost, scheduling, storage, and grid interaction than around explicit occupant modeling, preference feedback, or broader human-centric metrics.

S4.8 Reinforcement learning: theory and applications in HEMS [8]

Table S4.43: RQ4 recoded audit table for Al-Ani & Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	HEMS scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Al-Ani and Das (2022)	3	1	2	2	1	The paper is a specialized review of reinforcement learning in home energy management systems. HEMS scope receives score 3 because home energy management is the declared and structuring focus of the review. Occupant-framing visibility receives score 1 because occupants appear mainly through comfort as an optimization objective, but the paper gives little direct treatment to occupants as behavioral agents, feedback sources, or personalization targets. Metric visibility receives score 2 because energy use, cost, demand response, storage, renewable generation, and occupant comfort are discussed as objectives or application concerns, although human-centric metrics do not structure the review. Metric-imbalance visibility receives score 2 because the paper is organized much more strongly around HEMS energy/cost/grid-facing applications than around occupant-centered outcomes. Gap-signal visibility receives score 1 because real applications and case-study maturity are noted as concerns, but occupant-specific human-centric gaps are not developed as a major future-work theme.

Table S4.44: RQ4 occupant-framing scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Al-Ani & Das (2022)	0	1	0	0	0	0

Table S4.45: RQ4 metric-visibility scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Al-Ani & Das (2022)	3	3	2	2	0	0	1	0

Table S4.46: RQ4 gap-signal scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Al-Ani & Das (2022)	0	0	0	0	1

Table S4.47: RQ4 summary table for Al-Ani & Das (2022), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Al-Ani & Das (2022)	Mostly energy- and cost-centric, with comfort as a co-objective	The review mainly frames occupants as users whose comfort may be considered within automated home energy management, but it gives very little direct attention to explicit occupant behavior models, feedback-driven adaptation, or personalization.	For RQ4, the paper is best interpreted as showing that RL-for-HEMS research is organized primarily around energy management, cost reduction, demand response, storage, and broader whole-home coordination, while human-centered treatment appears mainly through comfort maintenance rather than through richer occupant-centric modeling or preference-aware control. This supports a clear imbalance in which comfort is present but secondary to broader energy-management objectives.

Table S4.48: One-line RQ4 answer for Al-Ani & Das (2022), after recoding.

Survey	RQ4 answer
Al-Ani and Das (2022)	Al-Ani & Das (2022) portray occupants mainly as users whose comfort may be preserved within automated home energy management, while the reviewed RL literature is organized much more strongly around energy use, cost reduction, demand response, storage, and whole-home coordination than around explicit occupant modeling, preference feedback, or personalization.

S4.9 Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges [9]

Table S4.49: RQ4 recoded audit table for Chatterjee & Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Dynamic thermal/HVAC scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Chatterjee and Khovalyg (2023)	3	3	3	3	3	The paper is a specialized review of reinforcement-learning-based HVAC control for creating dynamic indoor thermal environments. Dynamic thermal/HVAC scope receives score 3 because dynamic indoor thermal environment control and RL-based HVAC control are the declared and structuring focus of the review. Occupant-framing visibility receives score 3 because occupants, thermal adaptation, comfort, health, thermal preference diversity, activity, clothing, expectations, physiology, productivity, and energy expenditure structure the review motivation. Metric visibility receives score 3 because the paper structurally centers energy, thermal comfort, and health/wellbeing as interacting objectives. Metric-imbalance visibility receives score 3 because it explicitly contrasts existing RL-HVAC studies, which mainly focus on energy reduction and sometimes thermal comfort, with the underdeveloped aim of creating genuinely dynamic indoor environment conditions. Gap-signal visibility receives score 3 because the lack of RL studies for dynamic indoor environments is one of the central gaps motivating the review.

Table S4.50: RQ4 occupant-framing scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Chatterjee & Khovalyg (2023)	2	3	0	1	2	3

Table S4.51: RQ4 metric-visibility scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Chatterjee & Khovalyg (2023)	3	2	0	3	3	0	3	2

Table S4.52: RQ4 gap-signal scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Chatterjee & Khovalyg (2023)	2	0	3	3	2

Table S4.53: RQ4 summary table for Chatterjee & Khovalyg (2023), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Chatterjee & Khovalyg (2023)	Mixed, but energy–thermal–health centered	The review mainly frames occupants as thermally adaptive users whose comfort, health, wellbeing, productivity, and energy expenditure are shaped by dynamic indoor temperature conditions. It is strongly motivated by human-centered thermal adaptation, but gives less direct attention to explicit feedback-based interaction or operational personalization inside the RL loop.	For RQ4, the paper is best interpreted as showing that this specialized RL-for-HVAC review is strongly motivated by human-centered concerns such as comfort, adaptation, health, wellbeing, and productivity. However, the underlying RL literature still focuses most heavily on energy savings and thermal-comfort maintenance, while genuinely dynamic indoor thermal environments and richer health/IEQ-centered control remain underdeveloped.

Table S4.54: One-line RQ4 answer for Chatterjee & Khovalyg (2023), after recoding.

Survey	RQ4 answer
Chatterjee and Khovalyg (2023)	Chatterjee & Khovalyg (2023) portray occupants mainly as thermally adaptive users whose comfort, health, wellbeing, productivity, and energy expenditure are affected by dynamic indoor temperature conditions. The paper shows that RL-HVAC research remains more developed around energy savings and thermal-comfort maintenance than around genuinely dynamic, health- and IEQ-centered indoor-environment control.

S4.10 Building energy management with reinforcement learning and model predictive control: A survey [10]

Table S4.55: RQ4 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad BEMS scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Zhang et al. (2022)	3	2	3	2	1	The paper is a broad survey of building energy management with reinforcement learning and model predictive control. Broad BEMS scope receives score 3 because BEMS is the declared and structuring application domain of the review. Occupant-framing visibility receives score 2 because occupants are discussed mainly through comfort requirements, variable occupancy conditions, and human interaction with the building, but occupants do not structure the paper as behavioral agents or feedback sources. Metric visibility receives score 3 because the review repeatedly covers energy consumption, electricity cost, peak reduction, load shifting, scheduling, user comfort, indoor temperature, humidity, and broader system-efficiency objectives. Metric-imbalance visibility receives score 2 because the energy/cost/control orientation is clearly stronger than the human-centric framing, although the paper is not organized as a metric-imbalance review. Gap-signal visibility receives score 1 because occupant-related limitations appear only weakly, mainly through generalization under changing conditions, while the strongest challenge discussion is technical rather than explicitly human-centered.

Table S4.56: RQ4 occupant-framing scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Zhang et al. (2022)	2	3	0	0	0	1

Table S4.57: RQ4 metric-visibility scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Zhang et al. (2022)	3	3	2	2	1	0	1	0

Table S4.58: RQ4 gap-signal scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Zhang et al. (2022)	0	0	0	1	0

Table S4.59: RQ4 summary table for Zhang et al. (2022), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Zhang et al. (2022)	Mixed, but energy–cost dominant	The review mainly frames occupants as comfort-constrained building users and as part of the external uncertainty context through variable occupancy conditions and human interaction with the building. It gives little direct attention to feedback-rich human interaction, explicit behavioral modeling, or personalization.	For RQ4, the paper is best interpreted as showing that RL- and MPC-based BEMS research is organized primarily around energy consumption, electricity cost, peak reduction, load shifting, and broader system efficiency, while human-centered treatment appears mainly through thermal comfort and acceptable indoor temperature/humidity ranges. This supports a clear imbalance in which comfort is present but secondary to broader energy-management objectives.

Table S4.60: One-line RQ4 answer for Zhang et al. (2022), after recoding.

Survey	RQ4 answer
Zhang et al. (2022)	Zhang et al. (2022) portray occupants mainly as comfort-constrained users embedded in a dynamic BEMS environment affected by occupancy and human interaction, while the surveyed literature remains organized much more strongly around energy consumption, electricity cost, peak reduction, load shifting, and broader control efficiency than around explicit occupant modeling, preference feedback, or personalization.

S4.11 Applications of reinforcement learning for building energy efficiency control: A review [11]

Table S4.61: RQ4 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Energy-efficiency scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Fu et al. (2022)	3	2	3	2	3	The paper is a broad review of reinforcement learning for building energy-efficiency control. Energy-efficiency scope receives score 3 because building energy-efficiency control is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because occupants are discussed mainly through comfort and safety requirements, comfort-violation penalties, and stochastic occupant behavior, but occupants do not structure the whole paper as feedback-giving or personalized agents. Metric visibility receives score 3 because the review repeatedly covers energy consumption, energy cost, thermal comfort, comfort-violation penalties, demand-response/grid objectives, and control-performance objectives. Metric-imbalance visibility receives score 2 because energy and cost objectives dominate while comfort appears mainly as a recurring co-objective or penalty term. Gap-signal visibility receives score 3 because occupant behavior is explicitly presented in the future-directions discussion as stochastic, complex, influential for equipment and HVAC operation, and a promising direction for RL-based building control.

Table S4.62: RQ4 occupant-framing scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Fu et al. (2022)	3	2	0	2	0	1

Table S4.63: RQ4 metric-visibility scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Fu et al. (2022)	3	2	2	2	0	0	1	0

Table S4.64: RQ4 gap-signal scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Fu et al. (2022)	0	0	0	3	1

Table S4.65: RQ4 summary table for Fu et al. (2022), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Fu et al. (2022)	Mostly energy-centric, with comfort as a recurring co-objective	The review mainly frames occupants as users whose comfort and safety must be preserved while building systems are optimized for energy savings and cost reduction. It also acknowledges occupant behavior as a stochastic and operationally important factor, but mainly as a future research challenge rather than as a richly developed modeling dimension in the surveyed literature.	For RQ4, the paper is best interpreted as showing that RL-for-building-energy-efficiency research is organized primarily around energy and cost objectives, while human-centered treatment appears mainly through thermal comfort maintenance, discomfort reduction, or comfort-violation penalties. Thus, the survey supports a clear imbalance in which comfort is present and important, but richer occupant-centered treatment remains thinner and more future-oriented.

Table S4.66: One-line RQ4 answer for Fu et al. (2022), after recoding.

Survey	RQ4 answer
Fu et al. (2022)	Fu et al. (2022) portray occupants mainly as comfort-constrained users within an energy-efficiency control problem, while the reviewed RL literature remains organized much more strongly around energy savings, cost reduction, and broader control performance than around explicit occupant modeling, feedback-rich interaction, or personalization. The paper nevertheless identifies stochastic occupant behavior as an important future direction for RL-based building control.

S4.12 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [12]

Table S4.67: RQ4 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	HEMS scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Latoń et al. (2024)	3	2	3	3	2	The paper is a specialized review of reinforcement learning and deep reinforcement learning for home energy management systems. HEMS scope receives score 3 because HEMS is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because users are discussed through habits, occupancy patterns, user comfort, user preferences, privacy, and satisfaction, but occupants are not developed as a full behavioral-modeling taxonomy. Metric visibility receives score 3 because the paper repeatedly covers energy optimization, cost reduction, demand response, scheduling, renewable integration, comfort, preferences, privacy, and user satisfaction. Metric-imbalance visibility receives score 3 because the review makes clear that cost, energy management, scheduling, demand response, and renewable integration dominate, while richer human-centered themes such as personalization, privacy, and satisfaction appear less frequently. Gap-signal visibility receives score 2 because the paper discusses user satisfaction, preferences, real-world case studies, scalability, privacy, and practical acceptance, but these do not become a fully structuring occupant-centered future-work agenda.

Table S4.68: RQ4 occupant-framing scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Latoń et al. (2024)	2	2	2	1	1	0

Table S4.69: RQ4 metric-visibility scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Latoń et al. (2024)	3	3	2	2	0	2	2	1

Table S4.70: RQ4 gap-signal scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Latoń et al. (2024)	1	2	0	2	2

Table S4.71: RQ4 summary table for Latoń et al. (2024), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Latoń et al. (2024)	Mostly energy- and cost-centric, with comfort as a recurring co-objective	The review mainly frames occupants as home users whose habits, occupancy patterns, preferences, and comfort requirements should be incorporated into HEMS operation, while also recognizing privacy concerns linked to user data. It gives limited direct attention to explicit behavioral modeling or strongly individualized control.	For RQ4, the paper is best interpreted as showing that RL/DRL-based HEMS research is organized primarily around cost reduction, energy management, scheduling, demand response, and renewable integration, while human-centered treatment appears mainly through comfort maintenance, user preferences, and user satisfaction. This supports a clear imbalance in which comfort and preference matter, but remain secondary to broader energy-management objectives.

Table S4.72: One-line RQ4 answer for Latoń et al. (2024), after recoding.

Survey	RQ4 answer
Latoń et al. (2024)	Latoń et al. (2024) portray occupants mainly as smart-home users whose habits, occupancy patterns, comfort needs, and preferences shape HEMS operation, but the surveyed RL/DRL literature remains organized much more strongly around cost reduction, energy optimization, scheduling, demand response, and renewable integration than around explicit occupant modeling, feedback-rich human interaction, or personalization.

S4.13 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [13]

Table S4.73: RQ4 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	SHEMS/storage scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Pinthurat et al. (2024)	3	2	3	3	2	The paper is a specialized review of reinforcement-learning-based approaches for smart home energy management systems with energy storage. SHEMS/storage scope receives score 3 because storage-integrated SHEMS is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because users appear through comfort needs, user preferences, user-related operating patterns, personalized automation, quality of life, privacy and security concerns, and user satisfaction in real-world validation, but occupants do not structure the paper as behavioral agents. Metric visibility receives score 3 because the review repeatedly covers energy efficiency, cost reduction, storage management, renewable integration, grid stability, user comfort, user preferences, and real-world satisfaction concerns. Metric-imbalance visibility receives score 3 because energy, cost, storage, renewable integration, and grid stability dominate the paper, while human-facing metrics remain secondary co-objectives. Gap-signal visibility receives score 2 because real-world validation, user satisfaction, privacy, security, and personalization-related concerns are discussed, but they do not become the main structuring future-work agenda of the review.

Table S4.74: RQ4 occupant-framing scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Pinthurat et al. (2024)	2	2	1	1	1	0

Table S4.75: RQ4 metric-visibility scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Pinthurat et al. (2024)	3	3	2	2	0	2	1	1

Table S4.76: RQ4 gap-signal scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Pinthurat et al. (2024)	1	0	0	1	2

Table S4.77: RQ4 summary table for Pinthurat et al. (2024), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Pinthurat et al. (2024)	Mostly energy- and cost-centric, with comfort as a recurring co-objective	The review mainly frames occupants as smart-home users whose preferences, comfort needs, and behavior-related conditions shape energy management, while also recognizing privacy-sensitive user data and user satisfaction in real-world validation. It gives limited direct attention to explicit behavioral modeling or richer feedback-based interaction.	For RQ4, the paper is best interpreted as showing that RL-based SHEMS with energy storages are organized primarily around energy efficiency, cost reduction, storage optimization, renewable integration, and grid stability, while human-centered treatment appears mainly through user comfort, user preferences, and user satisfaction. This supports a clear imbalance in which human-facing concerns are present but remain secondary to broader energy-management objectives.

Table S4.78: One-line RQ4 answer for Pinthurat et al. (2024), after recoding.

Survey	RQ4 answer
Pinthurat et al. (2024)	Pinthurat et al. (2024) portray occupants mainly as smart-home users whose preferences, comfort needs, and behavior-related operating conditions shape energy management, but the reviewed RL-based SHEMS literature remains organized much more strongly around energy efficiency, cost reduction, storage optimization, renewable integration, and grid stability than around explicit occupant modeling, feedback-rich interaction, or broader human-centric metrics.

S4.14 A review of reinforcement learning for controlling building energy systems from a computer science perspective [14]

Table S4.79: RQ4 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Computer-science/BES scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Weinberg et al. (2023)	3	1	2	2	0	The paper is a specialized computer-science-oriented review of reinforcement learning for controlling building energy systems. Computer-science/BES scope receives score 3 because the review is structurally organized around computational challenges for RL in BES control, including single buildings, building clusters, and multi-agent aspects. Occupant-framing visibility receives score 1 because occupants appear mainly through thermal-comfort requirements in reward design, not as behavioral agents, feedback sources, or personalization targets. Metric visibility receives score 2 because energy use, cost, and thermal-comfort violations are discussed as relevant reward components, but human-centric metrics do not structure the review. Metric-imbalance visibility receives score 2 because the paper is clearly more oriented toward RL methodology, scalability, sample efficiency, transfer, and system design than toward occupant-centered outcomes. Gap-signal visibility receives score 0 because the paper’s future directions are computer-science and control challenges rather than human-centric occupant or comfort gaps.

Table S4.80: RQ4 occupant-framing scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Weinberg et al. (2023)	0	2	0	0	0	0

Table S4.81: RQ4 metric-visibility scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Weinberg et al. (2023)	3	2	1	2	0	0	0	0

Table S4.82: RQ4 gap-signal scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.					
Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Weinberg et al. (2023)	0	0	0	0	0

Table S4.83: RQ4 summary table for Weinberg et al. (2023), after recoding.			
Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Weinberg et al. (2023)	Mostly energy- and methodology-centric, with comfort as a reward constraint	The review mainly frames occupants as people whose thermal comfort should not be violated while building energy systems are optimized. It gives little to no direct attention to occupant feedback, explicit behavioral representation, personalization, or broader IEQ outcomes.	For RQ4, the paper is best interpreted as showing that this computer-science-oriented RL-for-BES review is organized primarily around energy control, cost, sample efficiency, transfer learning, scalability, multi-agent coordination, and RL system design, while human-centered treatment is narrowed largely to thermal-comfort violations in reward design.

Table S4.84: One-line RQ4 answer for Weinberg et al. (2023), after recoding.	
Survey	RQ4 answer
Weinberg et al. (2023)	Weinberg et al. (2023) portray occupants mainly through thermal-comfort violations in BES reward design, while the review remains organized much more strongly around energy use, cost, scalability, transfer learning, multi-agent coordination, and reinforcement-learning methodology than around explicit occupant modeling, feedback-rich interaction, personalization, or broader IEQ outcomes.

S4.15 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [15]

Table S4.85: RQ4 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	HVAC scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Sierla et al. (2022)	3	2	3	3	3	The paper is a specialized review of reinforcement learning applications to HVAC control. HVAC scope receives score 3 because HVAC control is the declared and structuring focus of the review. Occupant-framing visibility receives score 2 because occupants are mainly represented through thermal-comfort requirements, indoor-environment variables, and a few user-request or occupancy-related signals, but occupants do not structure the review as behavioral agents. Metric visibility receives score 3 because the paper structurally analyzes indoor-environment target variables, including temperature, humidity, and air quality, alongside energy and cost objectives. Metric-imbalance visibility receives score 3 because the paper explicitly shows that most reviewed works consider thermal comfort while only a small minority use standard comfort metrics and the great majority ignore air quality. Gap-signal visibility receives score 3 because the underrepresentation of indoor air quality and richer IEQ variables is a structural gap signal in the paper’s synthesis.

Table S4.86: RQ4 occupant-framing scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Sierla et al. (2022)	0	3	1	0	0	2

Table S4.87: RQ4 metric-visibility scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Sierla et al. (2022)	3	2	0	3	3	1	1	0

Table S4.88: RQ4 gap-signal scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Sierla et al. (2022)	0	1	3	0	0

Table S4.89: RQ4 summary table for Sierla et al. (2022), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Sierla et al. (2022)	Mixed, but energy–thermal-comfort dominant	The review mainly frames occupants as users whose indoor thermal conditions must remain acceptable, with comfort represented primarily through temperature and only secondarily through humidity or indoor air quality. It gives much less direct attention to occupants as explicit behavioral models, preference-rich agents, or personalization targets.	For RQ4, the paper is best interpreted as showing that RL-for-HVAC research is strongly centered on the trade-off between energy or cost objectives and temperature-based thermal comfort, while richer human-centered dimensions remain thinner. Indoor air quality and humidity are present but much less common, and explicit occupant feedback or personalization appears only sporadically.

Table S4.90: One-line RQ4 answer for Sierla et al. (2022), after recoding.

Survey	RQ4 answer
Sierla et al. (2022)	Sierla et al. (2022) portray occupants mainly as thermal-comfort-constrained users within HVAC control, while the reviewed RL literature remains organized much more strongly around energy or cost optimization plus temperature-based comfort than around indoor air quality, richer IEQ outcomes, explicit occupant modeling, or personalization.

S4.16 A review of deep reinforcement learning for smart building energy management [16]

Table S4.91: RQ4 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad SBEM scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Yu et al. (2021)	3	2	3	2	1	The paper is a broad review of deep reinforcement learning for smart building energy management. Broad SBEM scope receives score 3 because smart building energy management is the declared and structuring application domain of the review. Occupant-framing visibility receives score 2 because occupants are discussed through user comfort, occupancy, occupant number, CO ₂ concentration, indoor environmental constraints, and selective feedback-based behavior-shaping examples, but occupants do not structure the paper as behavioral agents or personalization targets. Metric visibility receives score 3 because the review repeatedly covers energy consumption, carbon emission, energy cost, user comfort, thermal comfort, indoor air quality, lighting comfort, occupant satisfaction, safety, and productivity. Metric-imbalance visibility receives score 2 because the review shows that energy and cost objectives dominate while human-centric metrics are present mainly through comfort and selected IEQ examples, but metric imbalance is not the paper’s main organizing analysis. Gap-signal visibility receives score 1 because broader human-centric gaps such as feedback-rich interaction, IEQ, behavioral realism, and real-world validation appear only weakly compared with the paper’s stronger technical and energy-management focus.

Table S4.92: RQ4 occupant-framing scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Yu et al. (2021)	2	3	1	1	0	2

Table S4.93: RQ4 metric-visibility scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Yu et al. (2021)	3	3	2	3	2	1	2	0

Table S4.94: RQ4 gap-signal scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Yu et al. (2021)	0	1	1	1	1

Table S4.95: RQ4 summary table for Yu et al. (2021), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Yu et al. (2021)	Mixed, but energy–cost dominant	The review mainly frames occupants as users whose thermal comfort and, in some works, indoor air quality and lighting conditions must be maintained, while also treating occupancy and occupant number as important sources of uncertainty. A smaller part of the review goes further by considering occupant satisfaction, productivity, and shaping occupant behavior through feedback-based recommendations.	For RQ4, the paper is best interpreted as showing that DRL-based SBEM research is still organized primarily around energy consumption, energy cost, and broader energy-management efficiency, with thermal comfort as the dominant human-centered metric. Broader human-centric dimensions such as air quality, lighting comfort, satisfaction, safety, productivity, and behavior shaping are present, but they are secondary and much less consistently developed across the reviewed literature.

Table S4.96: One-line RQ4 answer for Yu et al. (2021), after recoding.

Survey	RQ4 answer
Yu et al. (2021)	Yu et al. (2021) portray occupants mainly as comfort-constrained and partly uncertainty-generating users within smart building energy management, while the reviewed DRL literature remains organized much more strongly around energy consumption, cost, and system coordination than around explicit occupant modeling or personalization, with thermal comfort as the main human-centered metric and broader concerns such as IAQ, lighting comfort, productivity, and user feedback appearing more selectively.

S4.17 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [17]

Table S4.97: RQ4 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.						
Survey	Real-building deployment scope	Occupant-framing visibility	Metric visibility	Metric-imbalance visibility	Gap-signal visibility	Recoded evidence justification
Kannari et al. (2025)	3	3	3	3	3	The paper is a specialized review of reinforcement-learning-based control and optimization in real buildings. Real-building deployment scope receives score 3 because real-world RL deployment is the declared and structuring focus of the review. Occupant-framing visibility receives score 3 because occupants are discussed through comfort, indoor environmental quality, health, building-use conditions, individual thermal-comfort differences, personalized comfort models, surveys, wearable sensing, user feedback, privacy, and ethical concerns. Metric visibility receives score 3 because the review covers energy consumption, cumulative energy price, thermal comfort, indoor environmental quality, humidity, air velocity, particulate pollution, personalized comfort, and feedback-related comfort evaluation. Metric-imbalance visibility receives score 3 because the paper shows that most real-world studies still simplify occupant comfort mainly through indoor temperature while broader comfort variables and personalized comfort approaches are less common and harder to scale. Gap-signal visibility receives score 3 because the paper explicitly identifies broader IEQ, personalized comfort, scalable human-centered validation, privacy, and the narrow focus on heating/cooling rather than holistic building management as important limitations of the real-building RL literature.

Table S4.98: RQ4 occupant-framing scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.						
Survey	Occupant as disturbance / uncertainty	Occupant as comfort constraint	Occupant as feedback / preference source	Occupant as explicit behavioral model	Personalization / individual differences	Health / IEQ / wellbeing framing
Kannari et al. (2025)	2	3	2	1	2	3

Table S4.99: RQ4 metric-visibility scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Energy / efficiency	Cost / economic objective	Demand response / grid objective	Thermal comfort	IAQ / IEQ / health	Preference / satisfaction feedback	Behavioral realism / occupant interaction	Personalization / individual differences
Kannari et al. (2025)	3	2	0	3	3	2	2	2

Table S4.100: RQ4 gap-signal scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Personalization / individual differences	Feedback-rich human interaction	Health / IEQ beyond thermal comfort	Behavioral realism / non-stationarity	Real-world human-centric validation
Kannari et al. (2025)	2	2	3	2	3

Table S4.101: RQ4 summary table for Kannari et al. (2025), after recoding.

Survey	Metric orientation	Dominant occupant framing	Main RQ4 consequence
Kannari et al. (2025)	Mixed, but energy–thermal dominant	The review mainly frames occupants as users whose comfort, indoor environmental quality, and sometimes individual thermal preferences must be preserved in real-building HVAC control. It goes further than many HVAC reviews by discussing feedback, wearable sensing, personalized comfort models, privacy, and ethical concerns, but these remain less mature than the stronger deployment pattern of energy optimization plus temperature-based comfort maintenance.	For RQ4, the paper is best interpreted as showing that real-building RL deployments are still dominated by energy or price optimization balanced against thermal comfort, with comfort often simplified through indoor temperature. Broader IEQ variables, individualized comfort, richer feedback, privacy-aware sensing, and scalable human-centered validation are acknowledged and important, but remain less mature and more weakly represented than the core energy–temperature trade-off.

Table S4.102: One-line RQ4 answer for Kannari et al. (2025), after recoding.

Survey	RQ4 answer
Kannari et al. (2025)	Kannari et al. (2025) portray occupants mainly as comfort- and IEQ-constrained users in real-building RL control, while actual deployments remain organized primarily around energy use or energy price plus temperature-based thermal comfort. IAQ, health, personalization, feedback-rich interaction, privacy-aware sensing, and scalable human-centered validation are explicitly recognized but remain less consistently realized in practice.

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